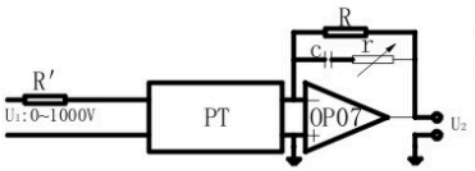
ZMPT101B(ZMPT107) voltage transformer operating guide

1、Wiring diagram



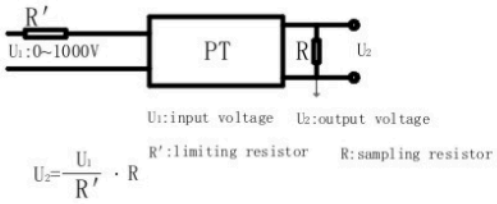


Figure Ⅰ **Figure** Ⅱ

2、Determination of maximum output rms voltage Umax:

Umax is decided by the AD peak voltage in the sampling loop in principle.

As for Bipolar AD, Umax= $\frac{Peakvoltag\ e}{\sqrt{2}}$

As for unipolar AD, Umax= $\frac{\text{peak}\ voltage}{2\sqrt{2}}$

for example:

As for ±5V AD, the maximum rms voltage of the transformer: Umax = $5V/\sqrt{2}$ **=3.53V**

As for 0~3.3V AD, the maximum rms voltage of

the transformer: $V_{max} = 3.3V/2\sqrt{2}$ **=1.16V**

3、Determination of input current-limiting resistor R′

Current-limiting resistor R′= $\frac{V}{I}$

V：Rated input voltage

I：Rated operating current（when Coil resistance is compared with current-limiting resistor R′, it can be ignored.）

ZMPT101B/ZMPT107 usually working at rated current:1~2mA. When Rated input voltage≤100V，Usually choosing the operating current I=2mA; When Rated input voltage≥220V，To reducing the resistor power，Usually choosing the operating current 1 mA ≤I≤2 mA.

for example: V=100V，I=2 mA，

$$R' = \frac{V}{I} = 50K\Omega$$

for example: V=220V，I=1.1mA，

$$R' = \frac{V}{I} = 200k\Omega$$

To improve reliability, the current-limiting resistor selected usually is greater than its 4times the rated power, and generally use a high

temperature coefficient metal film resistor.

4、Determination of the sampling resistor R

$$R=\frac{V_{output\max}}{I}=\frac{V_{output\max}}{V_{input\max}}\bullet R$$

for example: $V_{output\max}$ **=3.53V，** $V_{input\max}$ **=120V，** R' **=50** $k\Omega$

R= $\frac{3.53}{120}$ **×50** $k\Omega$ **=1.471** $k\Omega$

Directions:

(1) Above formula is also suitable for the two ways of active and passive output .

(2)when selecting the sampling resistor，Resistor should not exceed : $V_{output\max}\bullet R'/V_{input\max}$

5、The advantages and disadvantages of the two wiring

(1)、Active output

Advantage：high precise, small phase error，high output voltage，strong load capacity。

As for unipolar AD，the positive input terminal of the op amp can plus a fixed benchmark reference voltage to solve.

Meanwhile，in order to simplify the line, Generally do not access the c and r which are for the phase compensation. If you need to compensate，Usually

use the software way.

Disadvantage：Line is a little more complicated.

(2)、Passive output

Advantage：Simple circuit，High precision.

Disadvantage：The output voltage has certain limitations，The greater the load resistance, the greater the phase difference

Typical testing data are as follows：